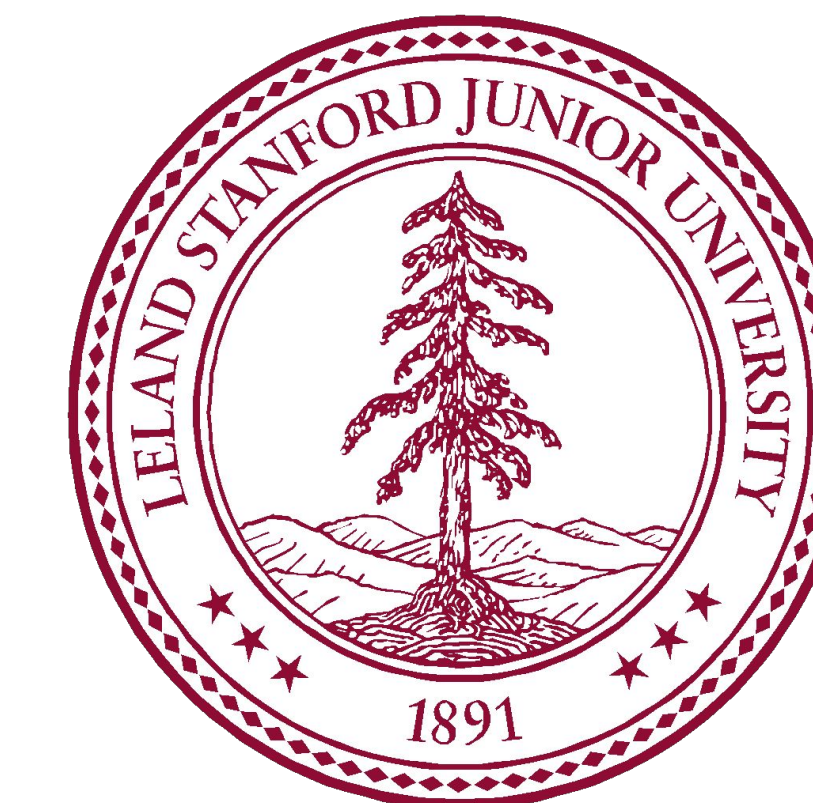




Multimodal Deep Learning to Predict Actionable Clinical Metrics from Unstructured Electronic Health Record and Imaging Data



Akshat Garg, Aaron Fanous, Roxana Daneshjou
Department of Biomedical Data Science
Stanford University, Stanford, CA 94305

Abstract

Clinical lab values from routine blood tests are key indicators of a patient's physiological status. Unlike mortality, which is delayed and influenced by treatment, lab values are updated frequently and directly guide clinical decisions.

Many current AI models predict long-term outcomes like mortality or readmission. However, these endpoints are often confounded by interventions and provide limited value in real-time care. In contrast, lab values offer numeric, interpretable signals that can inform medical decision-making.

We developed a multimodal deep learning model to predict structured lab values using only unstructured clinical notes, radiology reports, and chest X-rays. To train the model, we used data from the MIMIC-III and MIMIC-CXR databases. Lab values were removed from the input text to prevent data leakage.

Our model uses a vision-language model (VLM) to fuse imaging and text. It predicts lab values such as creatinine, hemoglobin, and white blood cell count. These outputs can also be used in counterfactual simulations of treatment effects.

In testing on 4,000 examples, the model achieved less than 5% absolute error in 93.4% of cases, with a mean absolute percentage error of 2.39%.

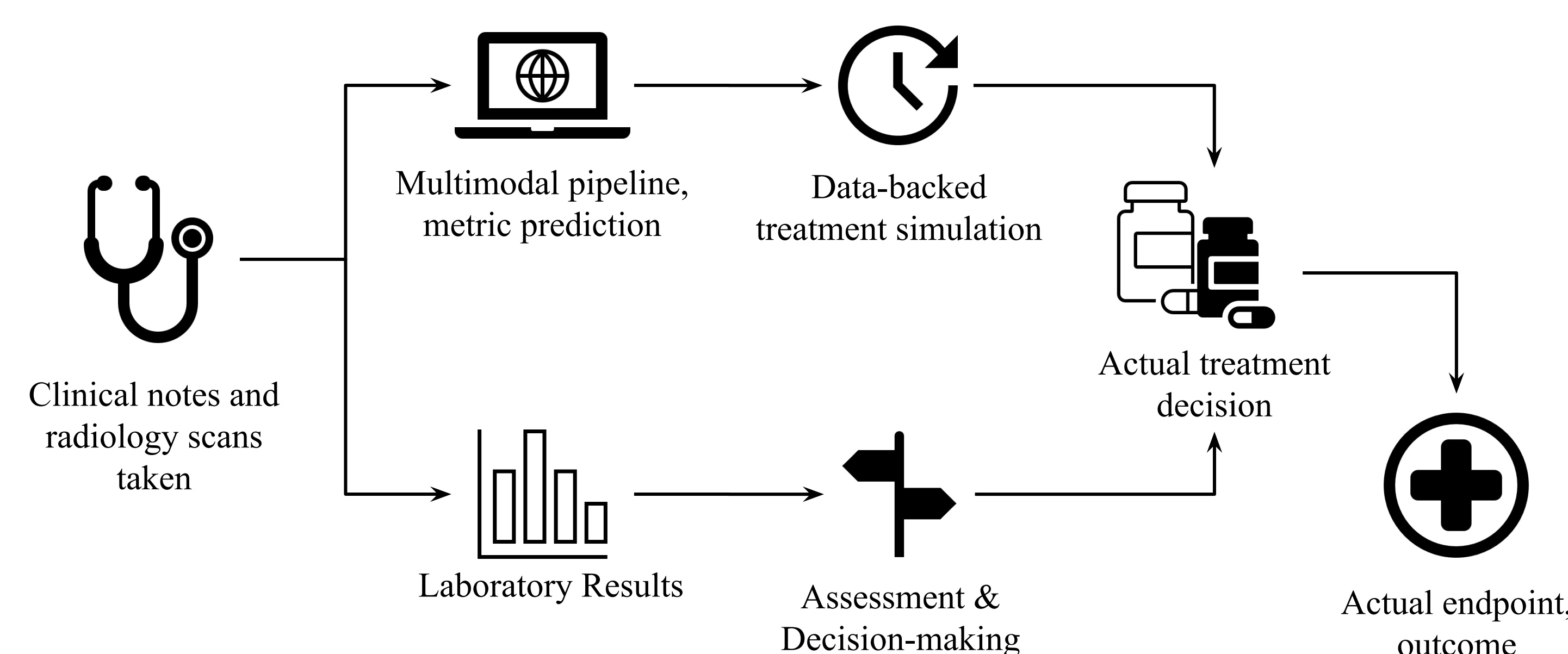
These results show that lab trends and the physiological states behind them can be inferred from unstructured clinical data. This approach enables real-time, interpretable decision support in dynamic care settings.

Introduction

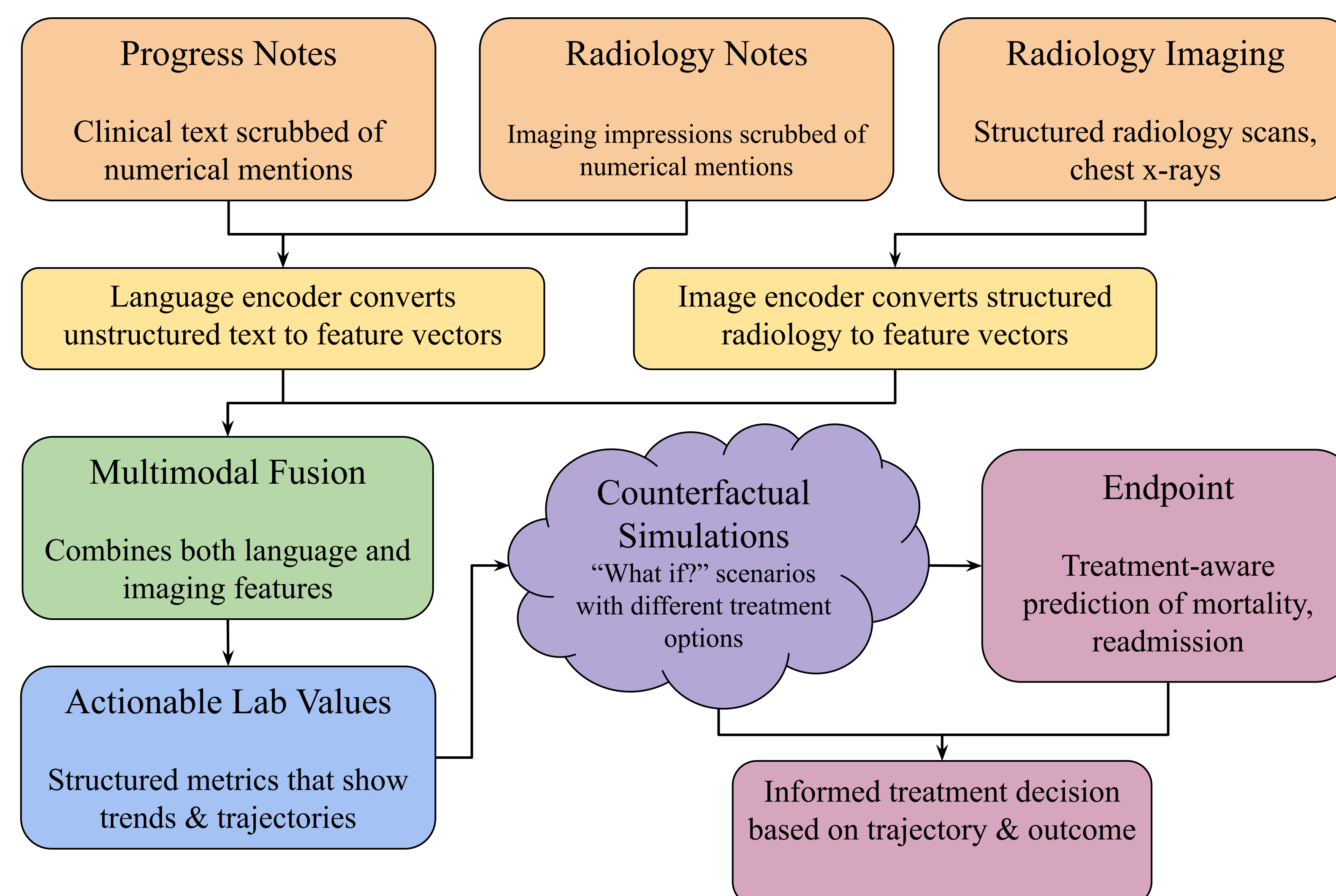
The Problem: While mortality is a common endpoint in clinical AI, it presents a late-stage outcome influenced by complex and evolving medical interventions, which are not fully addressed by those unfamiliar with the adaptivity of clinical decision making.

Our solution: align model outputs with the needs of physicians in the decision-making process. To best meet these needs, we geared the model to output metrics as indicators of patient status.

Proposed vs. Traditional Treatment Pipeline



Methods & Materials



The multimodal deep learning pipeline we created included the following components:

- Data Collection: Paired ICU patient data from **MIMIC-III** and **MIMIC-CXR**, progress/radiology notes and frontal chest x-rays
- Encoding
 - Unstructured text was encoded using a **transformer-based language model**
 - X-rays were processed through a pretrained **convolutional encoder**
- Prediction targets: components of **CBC** (Complete Blood Count) and **BMP** (Basic Metabolic Panel)
 - Often delayed, non-actionable tests in medical environment

Results

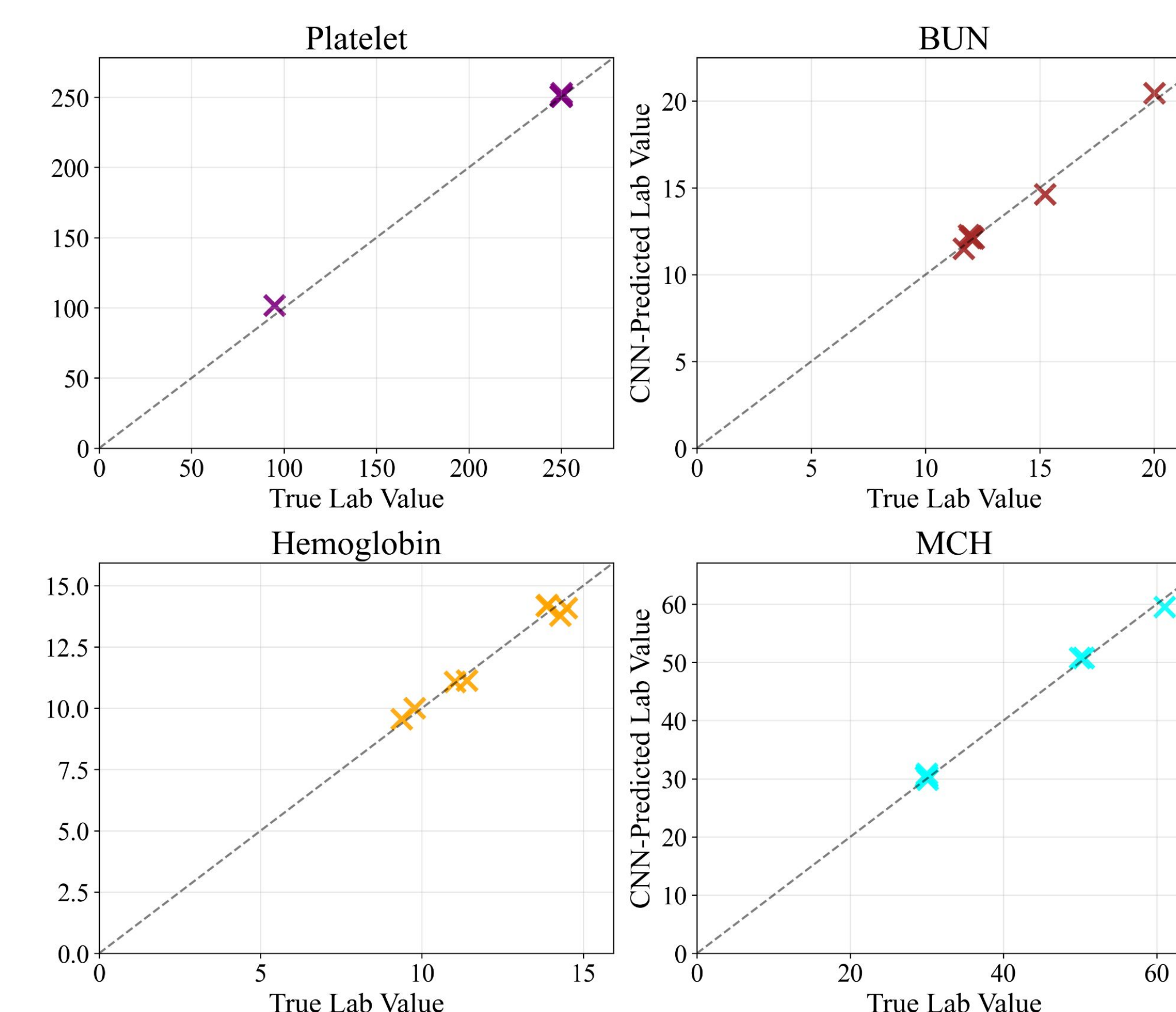


Figure 1. Testing using a simpler convolutional neural network (CNN) on **n=4000** samples was done to measure standard error in model predictions, among all the components of the CBC and BMP lab tests. The scatterplots show true vs. predicted values, with error under 5% at a 93.4% rate, and an **average error of 2.39%**.

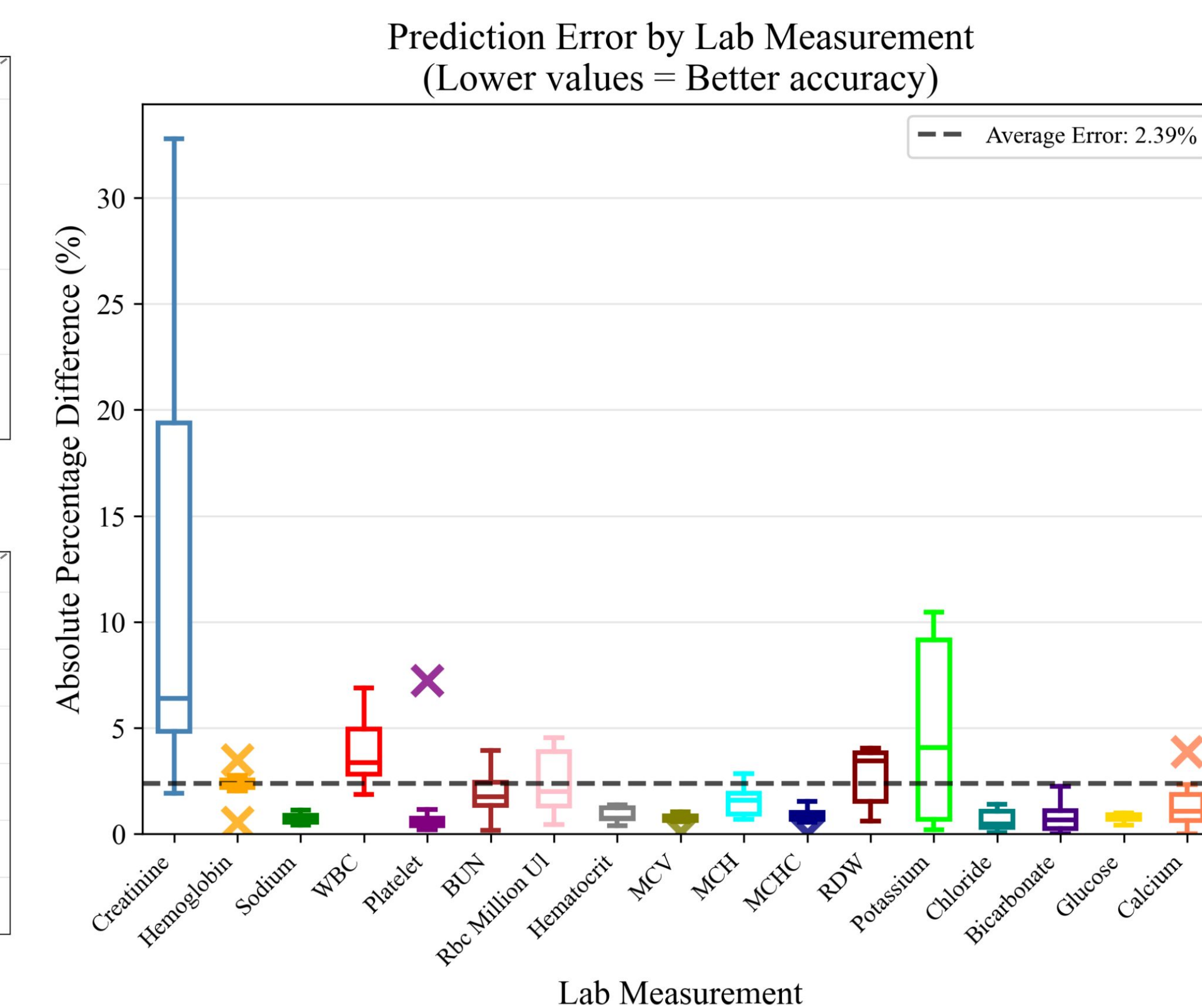


Figure 2. Across true vs. predicted comparisons, the error percentage was calculated. The boxplots illustrate **error percentages** to highlight consistency across physiological states. The plot displays **overall consistency** in predictions, with high outlier percentages in certain high-variance lab measurements, namely creatinine & potassium.

Conclusions

Based on the displayed accuracy, our hypothesis is supported and there is evidence toward the viability of predicted lab metrics in medical contexts.

Future Directions:

- Experimentation with model results and **generalization** with real-world data
 - Development of **simulations** for a more comprehensive set of lab metrics, treatments
 - Inference through **causal learning** to determine correlations with endpoints and metrics
 - Data **augmentation** with physiological accuracy after causal inference
- Further model development and causal inference, paired with new availability of diverse data, makes multimodal prediction increasingly feasible in medical contexts, for both the prediction and simulation of physiological trends.

Acknowledgements

This research was made possible thanks to the Stanford Institutes of Medicine Research (SIMR) Program, funding from the Strober Family Scholars Fund, and the generous support from members of the Daneshjou Lab at Stanford University.